

Sara Klaasen

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Education

PhD	ETH Zurich , Geophysics - Eidgenössische Technische Hochschule Zürich (ETH Zürich, Swiss Federal Institute of Technology Zurich) - ETH medal	Zurich, Switzerland 2020 – 2025
Msc	ETH Zurich , Geophysics	Zurich, Switzerland 2018 – 2020
Bsc	University College Roosevelt (UCR) , Liberal Arts and Sciences - Magna cum Laude - Major in Earth sciences, biology, mathematics and physics	Middelburg, Netherlands 2014 – 2017

Experience

Australian National University , Postdoctoral researcher Researching DAS projects in New Zealand. Imaging Taupo Volcano.	Canberra, Australia 2026 – present
ETH Zurich , Postdoctoral researcher	Zurich, Switzerland 2025 – 2026
Sport, Recreatie en Onderwijsvoorzieningen (SRO) , Admin Online administration	Haarlem, Netherlands 2017 – 2017
Provinciale Zeeuwse Courant (PZC) , Online editor - Online content management of the provincial newspaper - Writing and editing news articles	Vlissingen, Netherlands 2015 – 2017

Awards

ETH Medal ETH Zurich	2026
Outstanding Student Presentation Award (OSPA) Seismological Society of America Photonics Meeting	2024

Grants

Swiss National Science Foundation (SNF) postdoc.mobility grant Swiss National Science Foundation	2025
Women-in-STEM grant Fix the Leaky Pipeline program, ETH Zürich	2021
VSBFonds grant VSBFonds	2018

Research collaborations

Australian Research Council (ARC) Centre of Excellence in Optical Microcombs for Breakthrough Science (COMBS) Associate investigator www.combs.org.au	2026 – present
Seismicity on Venus: Prediction and Detection ECR member teams.issibern.ch/seismicityonvenus	2022 – 2024

Teaching experience

Data Analysis and Visualisation with Python in Earth Sciences Teaching assistant	2022 – 2025
Geophysical field lab Teaching assistant	2021 – 2023
Dynamische Erde I Guest lecturer	2020 – 2023

Research Output

Book Chapters

The leading edge of volcanology: Advances and innovations in monitoring and interpreting volcanic activity Michael P Poland, Elske de Zeeuw-van Dalfsen, Koki Aizawa, Sara Klaasen , Fiona D'Arcy, Emma J Nicholson, Kye Anderson Elsevier	2027
Fiber optic sensing for volcano monitoring and imaging volcanic processes Philippe Jousset, Gilda Currenti, Shane Murphy, Eva Eibl, Corentin Caudron, Lise Retailleau, Christopher Wollin, Sara Klaasen , Takeshi Nishimura, Andreas Fichtner, Zack Spica, Arnaud Lemarchand, Charlotte Krawczyk Springer doi.org/10.1007/978-3-031-86841-2_17	2025

Journal Articles

Submarine seismicity monitoring with distributed acoustic sensing near Santorini and Kolumbo Volcano Sara Klaasen , Thomas Hudson, Sanchit Sachdeva, Paraskevi Nomikou, Andreas Fichtner Seismica doi.org/10.26443/seismica.v5i1.1845	2026
Seismic constraints on glacier density Ariane Lanteri, Scott Keating, Lars Gebraad, Sara Klaasen , Marta Pienkowska-Cote, Olaf Eisen, Andrea Zunino, Kristín Jónsdóttir, Coen Hofstede, Dimitri Zigone, Andreas Fichtner Scientific Reports doi.org/10.1038/s41598-025-26440-z	2025
The Seismogenic Thickness of Venus Julia Maia, Ana-Catalina PLesá, Iris van Zelst, Richard Ghail, Anna Gülcher, Mark Panning, Sven Peter Näsholm, Barbara de Toffoli, Anna Horleston, Krystyna Smolinski, Sara Klaasen , Robert Herrick, Raphael Garcia Journal of Geophysical Research: Planets doi.org/10.1038/s41598-025-26440-z	2025
Towards a widely applicable earthquake detection algorithm for fibreoptic and hybrid fibreoptic-seismometer networks Thomas Hudson, Sara Klaasen , Olivier Fontaine, Conor Bacon, Kristin Jonsdottir, Andreas Fichtner Geophysical Journal International doi.org/10.1093/gji/ggae459	2025
An illustrated guide to: Distributed and integrated fibre-optic sensing in seismology Andreas Fichtner, Fabian Walter, Patrick Paitz, Sara Klaasen , Daniel Bowden, Sebastian Noe, Nils Müller, Dominik Husmann, Jacques Morel Earthquake Sciences dx.doi.org/10.1016/j.eq.s.2024.09.006	2024

- Seismic wave detectability on Venus using ground deformation sensors, infrasound sensors on balloons and airglow imagers 2024
 Raphael Garcia, Iris van Zelst, Taichi Kawamura, Sven Peter Näsholm, Anna Horleston, **Sara Klaasen**, Maxence Lefevre, Celine Marie Solberg, Krystyna Smolinski, Ana-Catalina Plesa, Quentin Brissaud, Julia Maia, Simon Stähler, Philippe Lognonné, Mark Panning, Anna Gülcher, Richard Ghail, Barbara de Toffoli
 Earth and Space Science
doi.org/10.1029/2024EA003670
- Modelling uncertainty in P-wave arrival-times retrieved from DAS data: case-studies from 15 fibre optic cables 2024
 Emanuele Bozzi, Nicola Piana Agostinetti, Andreas Fichtner, **Sara Klaasen**, Arantza Ugalde, Biondo Biondi, S Yuan, Torsten Dahm, Marius Isken, Patrick Paitz, Fabian Walter, Alain Baird, Carlos Becerril, Takeshi Nishimura, Junzhu Shen, Tieyuan Zhu, Gilberto Saccorotti
 Geophysical Journal International
doi.org/10.1093/gji/ggae364
- Complex spatial distribution of onset amplitude and waveform correlation: case studies from different DAS experiments 2024
 Emanuele Bozzi, Gilberto Saccorotti, Nicola Piana Agostinetti, Carlos Becerril, Andreas Fichtner, **Sara Klaasen**, Takeshi Nishimura, Junzhu Shen, Fabian Walter, Tieyuan Zhu
 Bulletin of geophysics and oceanography
doi.org/10.4430/bgo00458
- Challenges in submarine fiber-optic earthquake monitoring 2024
 Jonas Igel, **Sara Klaasen**, Sebastian Noe, Paraskevi Nomikou, Konstantinos Korantzalos, Andreas Fichtner
 Journal of Geophysical Research: Solid Earth
doi.org/10.1029/2024JB029556
- Subglacial volcano monitoring with fibre-optic sensing: Grímsvötn, Iceland 2023
Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristín Jónsdóttir, Lars Gebraad, Patrick Paitz, Andreas Fichtner
 Volcanica
doi.org/10.30909/vol.06.02.301311
- Fiber-optic observation of volcanic tremor through floating ice sheet resonance 2022
 Andreas Fichtner, **Sara Klaasen**, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Patrick Paitz, Kristin Jonsdottir
 The Seismic Record
doi.org/10.1785/0320220010
- Distributed acoustic sensing in volcano-glacial environments — Mount Meager, British Columbia 2021
Sara Klaasen, Patrick Paitz, Nadja Lindner, Jan Dettmer, Andreas Fichtner
 Journal of Geophysical Research: Solid Earth
doi.org/10.1029/2021JB022358
- Detecting Seismic Events with Computer Vision: Applications for Fiber-Optic Sensing 2021
 Solvi Thrastarson, Robert Torfason, **Sara Klaasen**, Patrick Paitz, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner
 ESS Open Archive
doi.org/10.1002/essoar.10509693.1
- First-Author Conference Proceedings**
- Modelling months of deformation during the Sundhnúkur eruptions on the Reykjanes Peninsula, Iceland 2026
Sara Klaasen, Daniel Bowden, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Jack Maxwell, Meghan Miller, Andreas Fichtner
 SSA optical seismology and the next era of seismic sensing

Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen , Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner IAVCEI General Meeting	2025
Full-Waveform Inversion of DAS data on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen , Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner IAVCEI General Meeting	2025
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen , Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner SSA General Meeting	2025
Moment Tensor inversion with Full-Waveform Inversion and Distributed Acoustic Sensing on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen , Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner EGU Copernicus Meetings	2024
Submarine volcano monitoring with fiber-optic cables at Santorini Sara Klaasen , Paraskevi Nomikou, Konstantinos Karantzas, Jonas Igel, Andreas Fichtner AGU Annual Meeting	2023
Going towards Full Waveform Inversion (FWI) with Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen , Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Jonas Igel, Andreas Fichtner AGU Annual Meeting	2023
Volcano monitoring and modelling using Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen , Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Sebastian Noe, Andreas Fichtner IUGG General Assembly	2023
Monitoring glacially covered volcanoes with Fibre-Optic Sensing Sara Klaasen , Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner IAVCEI General Meeting	2023
Volcanic Event Characterization with Fiber-Optic Sensing Sara Klaasen , Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner AGU Annual Meeting	2022
Fibre-Optic Sensing for Volcano Monitoring on Grímsvötn, Iceland Sara Klaasen , Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner EGU General Assembly	2022
Distributed Acoustic Sensing on Volcanoes: Grímsvötn and Fagradalsfjall, Iceland Sara Klaasen , Solvi Thrastarson, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Patrick Paitz, Andreas Fichtner AGU Annual Meeting	2021
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen , Patrick Paitz, Jan Dettmer, Andreas Fichtner SSA General Meeting	2021

Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen , Patrick Paitz, Jan Dettmer, Andreas Fichtner EGU General Assembly	2021
Distributed Acoustic Sensing for the Exploration of the Mount Meager Volcanic Complex, British Columbia, Canada Sara Klaasen , Andreas Fichtner, Patrick Paitz, Jan Dettmer AGU Annual Meeting	2020

Invited Talks

Keck observatory workshop SSA optical seismology and the next era of seismic sensing, Kona Pre-conference workshop	2026
Frontiers in Fiber-Optics Sensing in Seismology IASPEI-IAGA, Lisbon Conference session	2025
ECR Keynote talk IAVCEI, Geneva Conference keynote	2025
Seminar Stanford University, San Francisco Seminar	2025
Seminar University of Washington, online Seminar	2024
Session: An innovative photonic vision of volcanoes and geothermal systems SSA Photonic Seismology, Vancouver Session	2024
Colloquium CSIRO, Perth Seminar	2023
Geophysics Weekly Meeting Australian National University, Canberra Seminar	2023
Seminar University of Canterbury, Christchurch Seminar	2023
Workshop on Distributed Acoustic Sensing AGU, online Pre-conference workshop	2021

Service

Convening conference sessions
IAVCEI, AGU

Peer-review
Seismica, SRL

Languages

Dutch
native

English

C2

German

B2

French

B1